



An-Najah National University

Faculty of Information Technology and Artificial Intelligence

Advanced Topics in Machine Learning Course

Facial Emotion Recognition using EfficientNet-B2 and CBAM Attention

A Deep Learning Report Based on the FER2013 Dataset

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Date: 23 May 2026

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1. Summary

This project presents an advanced deep learning solution for Facial Emotion Recognition (FER) using the FER2013 dataset from Kaggle. The goal is to classify facial images into seven emotion categories: angry, disgust, fear, happy, neutral, sad, and surprise. A complete machine learning pipeline was built, starting with exploratory data analysis and dataset statistics, followed by preprocessing, augmentation, balanced sampling, model design, training, evaluation, explainability, and inference calibration. The proposed architecture uses an ImageNet-pretrained EfficientNet-B2 backbone enhanced with a Convolutional Block Attention Module (CBAM) and a custom classification head. The training process used class-weighted loss, WeightedRandomSampler, AdamW optimization, OneCycleLR scheduling, label smoothing, mixed precision, and early stopping. The final model achieved 63.86% test accuracy and 0.6170 Macro-F1. Temperature scaling reduced calibration error from 0.1049 to 0.0610, improving confidence reliability.

2. Introduction

2.1 Motivation

Facial expressions are one of the most natural forms of non-verbal communication. People often express feelings such as happiness, sadness, anger, fear, surprise, and disgust through small changes around the eyes, eyebrows, mouth, and cheeks. Automatically recognizing these expressions is an important research problem because it supports systems that need to understand human affective states, such as human-computer interaction platforms, online learning environments, mental health monitoring tools, customer feedback systems, driver monitoring systems, and assistive technologies.

The problem addressed in this project is the classification of facial images into emotion categories using deep learning. This task is challenging because facial expressions can be subtle, ambiguous, and highly variable between individuals. In addition, real-world images may contain differences in lighting, head pose, face shape, image quality, and occlusion. The FER2013 dataset also contains a strong class imbalance: the happy class is the majority class while disgust is severely underrepresented. Without careful design, a model may become biased toward majority emotions and perform poorly on rare but important classes.

This work is important because a reliable emotion recognition model should not only produce a predicted class, but should also provide interpretable and trustworthy outputs. Therefore, this project goes beyond training a standard classifier. It includes data analysis, imbalance handling, attention-based model design, Grad-CAM explainability, and confidence calibration. These components make the solution more suitable for an advanced machine learning setting because the model is evaluated not only by accuracy, but also by fairness across classes, error patterns, interpretability, and confidence reliability.

2.2 Background

Facial Emotion Recognition is commonly treated as a supervised image classification problem. Traditional approaches relied on handcrafted features such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), or geometric facial landmarks. These methods can work in controlled settings, but they often struggle when facial expressions vary naturally. Deep learning, especially Convolutional Neural Networks (CNNs), improved FER by learning hierarchical features directly from image pixels. Early layers capture edges and textures, while deeper layers learn more semantic facial regions such as eyes, eyebrows, nose, and mouth.

In this project, transfer learning was used through EfficientNet-B2 pretrained on ImageNet. Transfer learning is useful because FER2013 is relatively small compared with large-scale image datasets; using pretrained features helps the model start from general visual representations instead of learning everything from scratch. EfficientNet is suitable because it balances depth, width, and input resolution efficiently. To further improve the model, CBAM attention was added after the backbone feature extractor. CBAM applies channel attention to decide which

feature maps are important and spatial attention to decide where the model should focus. This is particularly useful for facial expression recognition because discriminative information is often localized around the mouth, eyes, and eyebrows.

The project also includes explainability and calibration. Grad-CAM was used to visualize which image regions influenced predictions, helping verify whether the model focused on meaningful facial areas. Temperature scaling was then applied to adjust predicted probabilities. Calibration is important because a classifier can be correct in class labels but still produce overconfident probabilities. A calibrated model gives confidence scores that better reflect actual correctness, which is valuable when predictions may be used in decision-support systems.

3. Dataset

3.1 Source

The dataset used in this project is FER2013, obtained from Kaggle: <https://www.kaggle.com/datasets/msambare/fer2013>. The dataset is widely used for facial emotion recognition and contains facial images grouped into seven emotional categories. In this project, the dataset was organized into train, validation, and test folders, each containing subfolders for the seven emotion classes.

3.2 Dataset Description

The dataset contains grayscale facial images. Since the model uses an ImageNet-pretrained EfficientNet-B2 backbone, images were converted into a 3-channel RGB-like format and resized to 224 x 224 pixels. The target variable is the emotion label, represented by one of seven classes: angry, disgust, fear, happy, neutral, sad, and surprise.

Split	Angry	Disgust	Fear	Happy	Neutral	Sad	Surprise	Total
Train	3,396	371	3,483	6,133	4,221	4,106	2,696	24,406
Validation	599	65	614	1,082	744	724	475	4,303
Test	958	111	1,024	1,774	1,233	1,247	831	7,178

The class distribution shows a major imbalance. The happy class is the majority class in all splits, while disgust is the smallest class. In the training split, happy contains 6,133 samples while disgust contains only 371 samples, giving a global imbalance ratio of approximately 16.53x. This imbalance can cause a model to learn majority classes more easily and ignore minority classes unless corrective techniques are used.

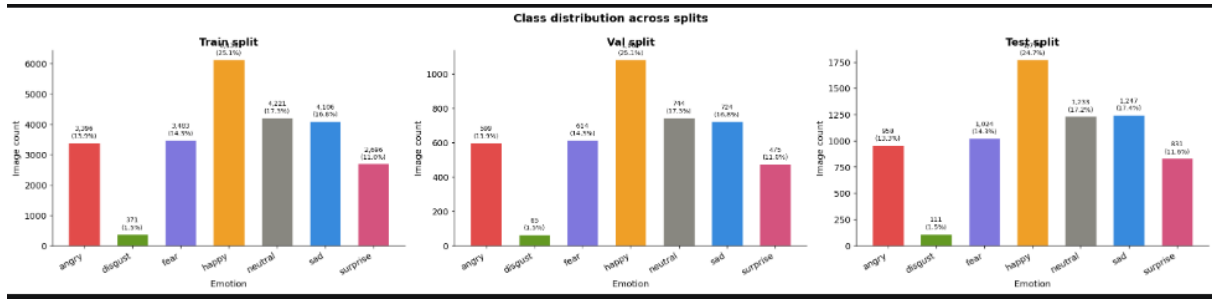


Figure 1. Class distribution across train, validation, and test splits.

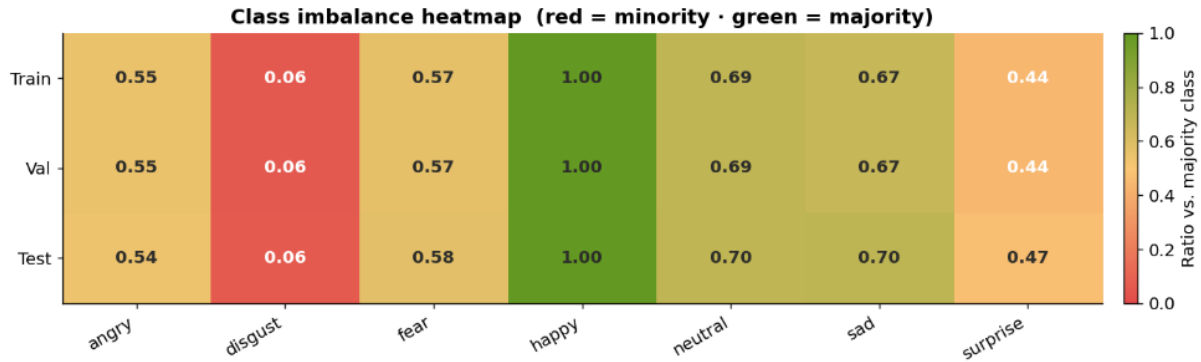


Figure 2. Class imbalance heatmap showing ratio against the majority class.

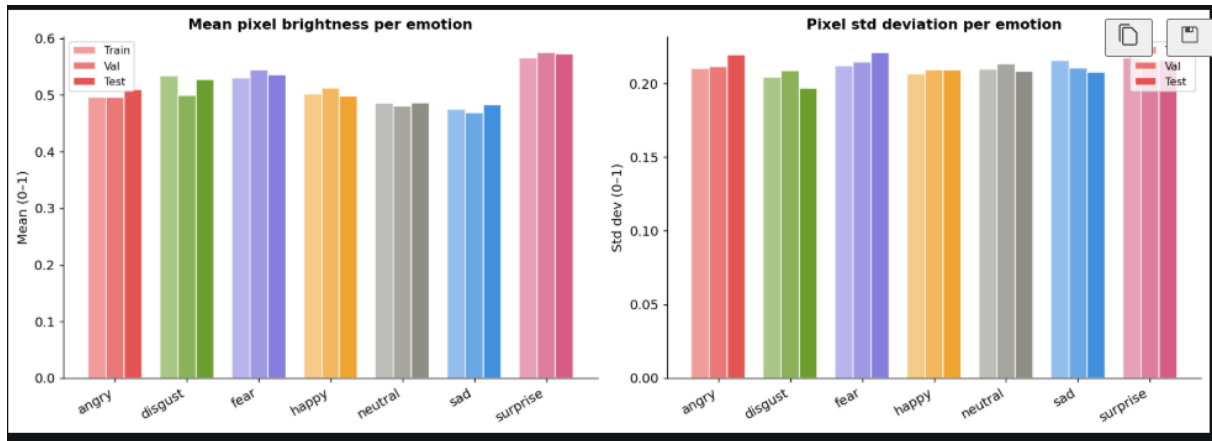


Figure 3. Pixel brightness and standard deviation statistics per emotion.

3.3 Feature Description

Feature/Element	Description
Image pixels	The input visual information representing a facial expression.
Emotion label	The supervised target class: angry, disgust, fear, happy, neutral, sad, or surprise.
Split	The dataset subset used for training, validation, or testing.
Brightness statistics	Mean pixel intensity used to understand global illumination patterns.

Pixel standard deviation	Variation in pixel values, useful for understanding contrast and image diversity.
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3.4 Preprocessing

Several preprocessing steps were applied before training. Images were loaded using a custom PyTorch Dataset class, resized to match the pretrained backbone input requirement, converted to tensors, and normalized using dataset-specific mean and standard deviation values. The EDA stage computed approximate normalization values of mean = 0.5050 and standard deviation = 0.2109. In the implemented data pipeline, normalization constants were stored and reused by later training phases.

Training data augmentation was applied to improve generalization. The training transform included resizing, random cropping, horizontal flipping, random rotation, color jitter, random grayscale conversion, Gaussian blur, tensor conversion, normalization, and random erasing. These augmentations simulate realistic variations such as head tilt, lighting changes, small occlusions, and facial position shifts. Validation and test transforms were kept lighter and deterministic to ensure fair evaluation.

Class imbalance was handled using two complementary techniques. First, WeightedRandomSampler was used during batch construction so that minority classes appeared more frequently during training. In a sanity check over one training epoch, all classes appeared at approximately 14% each, which means the sampler successfully balanced the batches. Second, class weights were passed to CrossEntropyLoss so that mistakes on minority classes received a larger penalty during optimization.

4. Methodology

4.1 Overall Approach

The methodology was organized into six phases. Phase 1 performed exploratory data analysis to understand the dataset distribution, imbalance, and pixel statistics. Phase 2 built the data pipeline, transformations, class weighting, and DataLoaders. Phase 3 designed the model architecture. Phase 4 trained the model and saved the best checkpoint based on validation Macro-F1. Phase 5 evaluated the model, analyzed errors, and produced Grad-CAM explanations. Phase 6 prepared the model for inference and applied temperature scaling for confidence calibration.

Phase	Purpose	Main Output
EDA	Analyze class counts, imbalance, and pixel statistics	Distribution plots and normalization statistics
Data Pipeline	Prepare datasets, transforms, and loaders	Balanced train loader and class weights
Architecture	Build EfficientNet-B2 + CBAM + FER head	Trainable FER model

Training	Optimize model parameters	Best checkpoint at epoch 40
Explainability	Analyze predictions and visual explanations	Confusion analysis and Grad-CAM
Inference/Calibration	Improve confidence reliability	Temperature-scaled model artifact

4.2 Model Architecture

The main model is based on EfficientNet-B2, pretrained on ImageNet. The backbone extracts visual feature maps from the input face image. The final feature map has 1408 channels and spatial dimensions of 7 x 7. A CBAM attention module is applied to this final feature map to refine the representation before classification. The model then uses global average pooling followed by a custom classification head.

The custom FER head contains two fully connected layers with batch normalization, ReLU activation, and dropout before the final seven-class output layer. The head structure is: 1408 -> 512 -> 256 -> 7. Dropout rates of 0.4 and 0.3 were used to reduce overfitting. The model outputs raw logits, and softmax probabilities are computed only during evaluation or inference.

Component	Configuration / Role
Backbone	EfficientNet-B2 pretrained on ImageNet
Attention	CBAM with channel and spatial attention
Feature size	Backbone output: 1408 x 7 x 7
Pooling	Global average pooling
Classifier head	FC(1408,512) -> BN -> ReLU -> Dropout(0.4) -> FC(512,256) -> BN -> ReLU -> Dropout(0.3) -> FC(256,7)
Output	Seven emotion logits
Parameters	8,804,971 total; 8,249,875 trainable; 555,096 frozen

4.3 CBAM Attention Mechanism

CBAM stands for Convolutional Block Attention Module. It contains two sequential attention mechanisms: channel attention and spatial attention. Channel attention answers the question: which feature maps are important? It uses average pooling and max pooling across spatial dimensions, passes them through a shared MLP, and generates a sigmoid gate for each channel. Spatial attention answers the question: where should the model focus? It compresses the feature map across channels using average and max operations, applies a convolution, and produces a spatial attention mask.

For facial emotion recognition, attention is useful because emotions are often expressed in local facial regions. For example, happiness is strongly connected to the mouth, surprise and fear often involve the eyes and eyebrows, and anger may involve the brow region. Adding CBAM encourages the network to focus on these discriminative regions rather than relying only on global image statistics.

4.4 Training Strategy

The training process used AdamW optimization with weight decay to improve generalization. A OneCycleLR scheduler was used to warm up the learning rate and then decay it smoothly, which often improves convergence stability. The base learning rate was $3e-4$ for the classification head and CBAM, while the unfrozen backbone layers used a lower learning rate of $3e-5$. This layer-wise learning rate strategy helps prevent catastrophic forgetting in the pretrained backbone while allowing task-specific layers to learn faster.

The model was trained for up to 50 epochs with early stopping patience of 10 epochs. Validation Macro-F1 was selected as the primary checkpointing metric because the dataset is imbalanced; Macro-F1 treats all classes equally, while accuracy may be dominated by majority classes. Label smoothing of 0.1 was applied to reduce overconfidence and improve robustness to ambiguous labels. Gradient clipping with a norm of 1.0 was used to stabilize optimization. Mixed-precision training was also applied to improve training speed and memory efficiency.

Hyperparameter	Value
Input size	224 x 224 x 3
Batch size	64
Optimizer	AdamW
Base learning rate	$3e-4$
Backbone learning rate	$3e-5$
Weight decay	$1e-4$
Label smoothing	0.1
Gradient clipping	1.0
Maximum epochs	50
Early stopping patience	10 epochs
Primary validation metric	Macro-F1

4.5 Evaluation Criteria

The model was evaluated using several metrics. Accuracy measures the percentage of correctly classified test images. Precision, recall, and F1-score were used to evaluate class-wise performance. Macro-F1 was emphasized because it gives equal weight to each emotion class and is therefore more suitable for imbalanced datasets. Confusion matrices were used to identify which emotion pairs were most frequently confused. Expected Calibration Error (ECE) was used to evaluate the reliability of predicted confidence scores.

Metric	Why it is appropriate for this project
Accuracy	Provides an overall measure of correct predictions.
Precision	Shows how reliable predictions are for each emotion class.
Recall	Shows how well each true emotion class is detected.
F1-score	Balances precision and recall, especially useful for class imbalance.
Macro-F1	Treats all classes equally and avoids majority-class dominance.

Confusion matrix	Reveals detailed error patterns between emotion pairs.
ECE	Measures whether confidence scores match actual correctness.

5. Results

5.1 Findings

The best checkpoint was obtained at epoch 40 with validation Macro-F1 = 0.6214. On the test set, the model achieved 63.86% accuracy and 0.6170 Macro-F1. These results indicate that the model learned meaningful facial expression features despite the imbalance and ambiguity of FER2013. The model performed best on happy and surprise, while fear and sad were more difficult.

Evaluation Result	Value
Best validation Macro-F1	0.6214 at epoch 40
Test loss	1.4960
Test accuracy	0.6386 (63.86%)
Test Macro-F1	0.6170
ECE before calibration	0.1049
ECE after temperature scaling	0.0610
Temperature value	0.8779
Calibration improvement	41.8% reduction in ECE

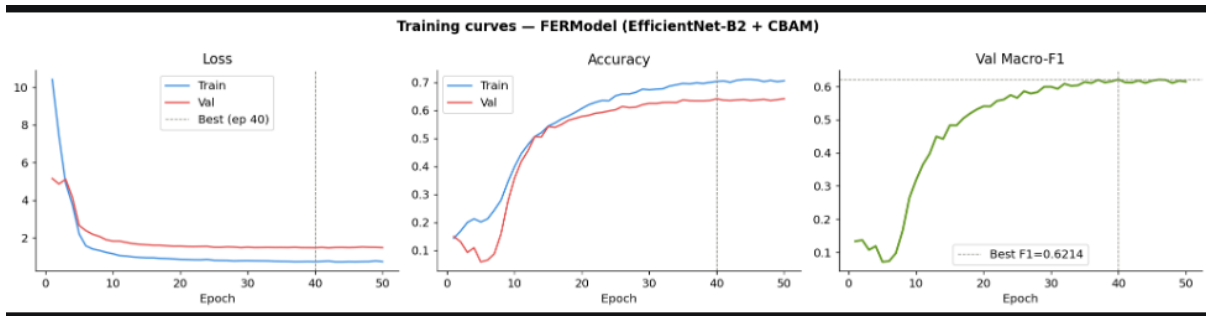


Figure 4. Training curves for loss, accuracy, and validation Macro-F1.

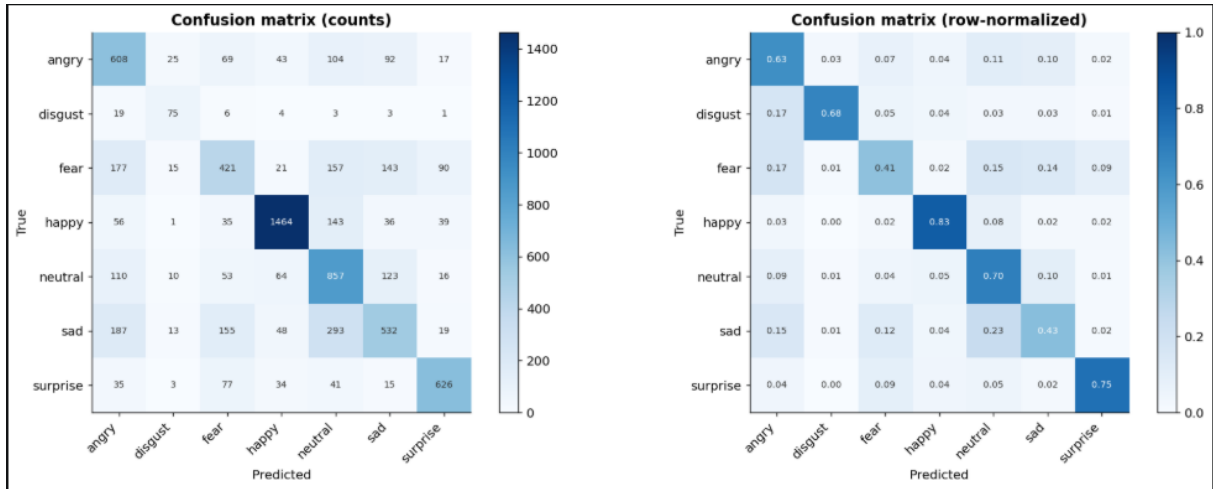


Figure 5. Confusion matrix in count and row-normalized forms.

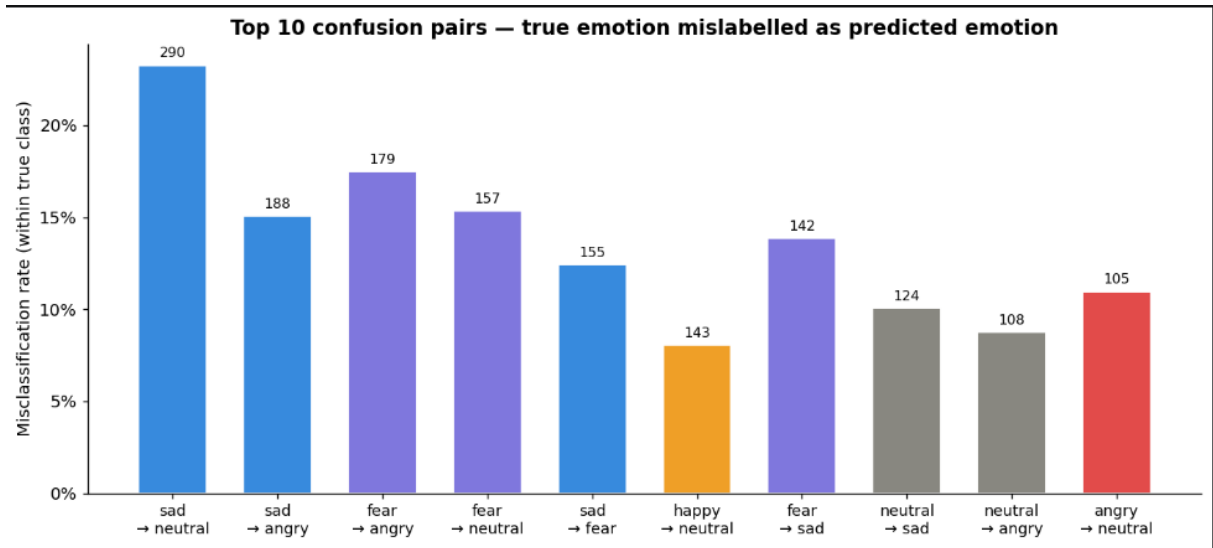


Figure 6. Top 10 most frequent confusion pairs.

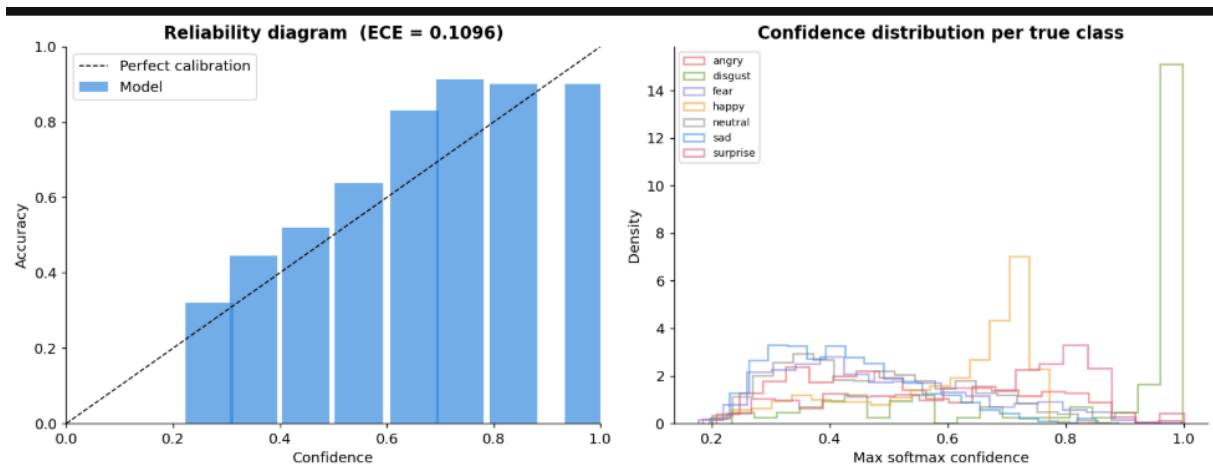


Figure 7. Reliability diagram and confidence distribution before calibration.

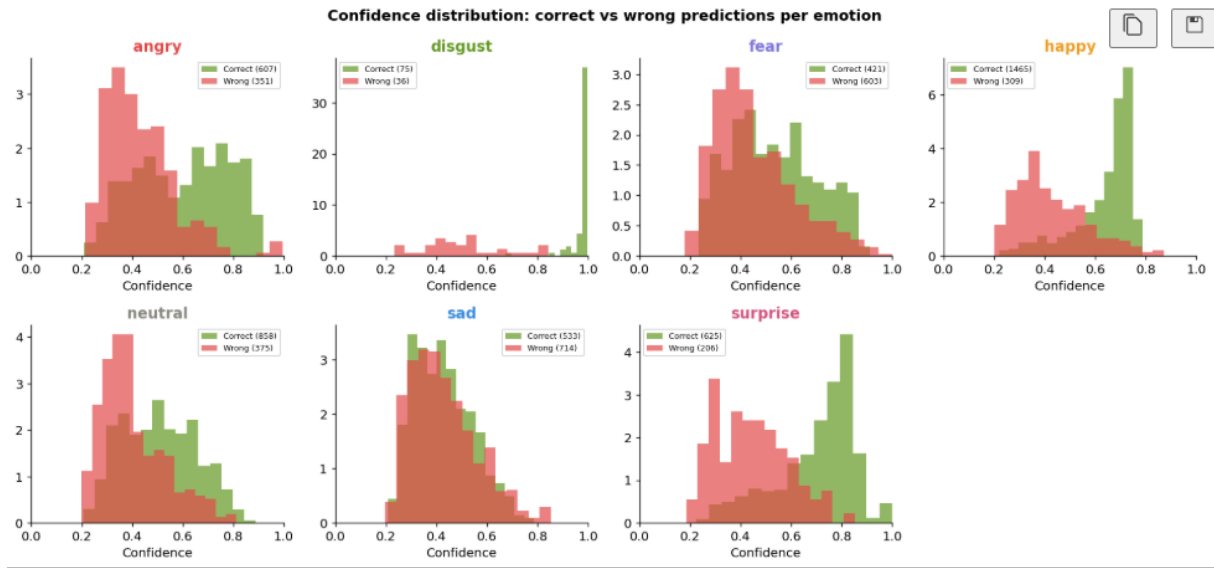


Figure 8. Confidence distributions for correct and wrong predictions per emotion.

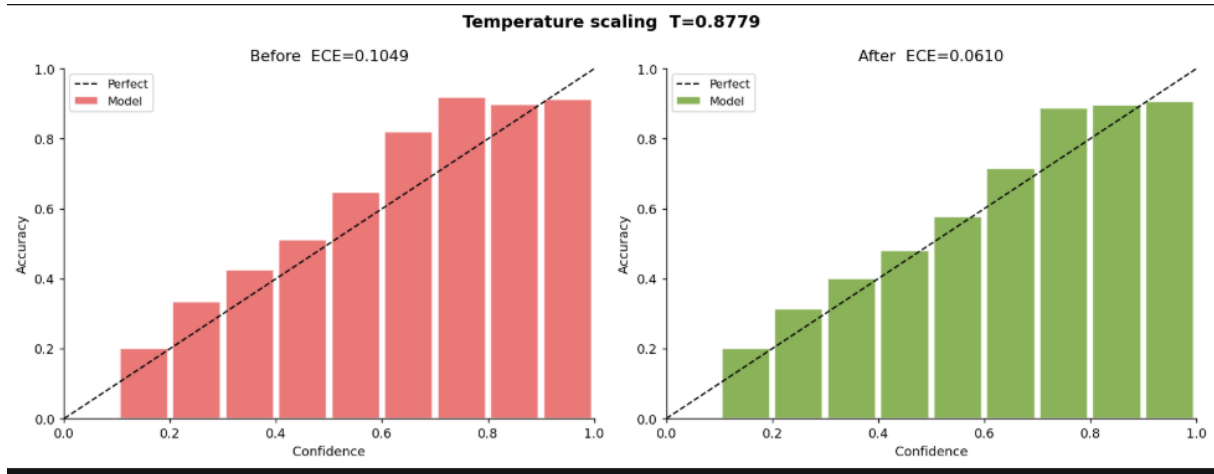


Figure 9. Temperature scaling comparison before and after calibration.

Emotion	F1-score	Interpretation
Happy	0.8488	Strongest class; many examples and visually distinctive smile patterns.
Surprise	0.7627	Strong performance; wide eyes and open mouth are discriminative.
Neutral	0.6059	Moderate performance; often confused with sad and angry.
Disgust	0.5929	Reasonable F1 despite very small sample count, supported by weighting and sampling.
Angry	0.5652	Moderate performance; confused with sad/fear/neutral in some cases.
Sad	0.4865	Difficult class; frequently

		confused with neutral and angry.
Fear	0.4574	Most difficult class; visually overlaps with angry, sad, and surprise.

The confusion matrix shows that happy is the easiest class, with a row-normalized recall of approximately 0.83. Surprise also performs strongly with approximately 0.75 recall. Neutral obtains around 0.70 recall. On the other hand, fear and sad have lower recall values of approximately 0.41 and 0.43, respectively. These results are expected because fear, sadness, anger, and neutral expressions can be visually subtle and sometimes share similar facial cues.

Confusion Pair	Count	Explanation
sad -> neutral	290	Sad expressions can be subtle and close to neutral facial appearance.
sad -> angry	188	Tension in eyebrows or mouth may resemble anger.
fear -> angry	179	Fear and anger may share strong eye/brow activation.
fear -> neutral	157	Mild fear expressions may appear neutral in low-resolution images.
sad -> fear	155	Both classes can involve similar mouth and eye patterns.
happy -> neutral	143	Weak smiles may be interpreted as neutral.
fear -> sad	142	Fear may be confused with negative affect classes.
neutral -> sad	124	Neutral faces with slight downturned features may appear sad.

5.2 Discussion

The results demonstrate that the proposed approach worked well for a difficult, imbalanced facial emotion recognition task. The use of transfer learning allowed the model to start with strong general visual features, while CBAM attention helped refine the final feature maps. Weighted sampling and class-weighted loss were important because the dataset was not balanced. Without these techniques, the model would likely overfit the majority happy class and ignore the minority disgust class.

The training curves show steady improvement in both accuracy and validation Macro-F1. Validation Macro-F1 increased significantly after the early epochs and reached its best value around epoch 40. The training accuracy continued to rise slightly after validation performance stabilized, which suggests that early stopping was useful to avoid unnecessary overfitting. The loss curve also shows that the model learned quickly during the first epochs and then gradually refined its decision boundaries.

The error analysis reveals that not all emotion classes are equally separable. Happy and surprise have strong visual signals and relatively high sample counts, so they achieved the highest F1-scores. Fear and sad were more challenging because their visual patterns overlap with other negative emotions. Additionally, FER2013 images are small and may contain ambiguous labels, which limits the maximum achievable performance. This is a realistic challenge in affective computing because emotions are subjective and can be expressed differently by different individuals.

The calibration results are an important contribution of this project. Before calibration, the model had an ECE of approximately 0.1049, indicating that its confidence scores were not perfectly aligned with actual accuracy. After applying temperature scaling with $T = 0.8779$, the ECE decreased to 0.0610, representing a 41.8% improvement. This means the final inference model produces more trustworthy confidence scores without changing the predicted labels. In practical applications, this is valuable because low-confidence predictions can be flagged for human review or treated with caution.

Grad-CAM explainability was included to inspect whether the model learned meaningful visual cues. The Grad-CAM analysis is especially useful for FER because the important regions should correspond to facial areas such as the mouth, eyes, cheeks, and eyebrows. If heatmaps focus on background regions instead of the face, the model may be learning shortcuts. In this project, Grad-CAM helped support the interpretation of model predictions and made the system more transparent.

5.3 Challenges and Limitations

- The dataset is imbalanced, especially for the disgust class, which has far fewer samples than happy.
- FER2013 images are low-resolution and some expressions are ambiguous or noisy.
- Fear, sad, angry, and neutral expressions overlap visually, producing many confusions.
- The model was evaluated on a fixed dataset and was not deployed in a real-time application.
- The model may not generalize equally to all ages, ethnicities, lighting conditions, cameras, or real-world environments.
- Calibration improved confidence reliability, but ECE is still not perfect and further calibration methods could be investigated.

6. Conclusion

This project developed a complete advanced machine learning pipeline for Facial Emotion Recognition using FER2013. The work covered dataset analysis, preprocessing, augmentation, imbalance handling, model architecture design, training, evaluation, explainability, and calibrated inference. The final architecture combined EfficientNet-B2, CBAM attention, and a

custom FER classification head. This design allowed the model to benefit from transfer learning while focusing on important facial regions.

The final model achieved 63.86% test accuracy and 0.6170 Macro-F1. The best-performing classes were happy and surprise, while fear and sad were the most challenging. Confusion matrix analysis showed that many errors occur between visually similar negative or neutral expressions. Temperature scaling improved confidence calibration by reducing ECE from 0.1049 to 0.0610. This makes the model's probabilities more reliable, which is important for real-world decision-support scenarios.

Overall, the project successfully addressed the FER classification problem using modern deep learning techniques. It also demonstrated that model performance should be evaluated beyond simple accuracy by considering imbalance, per-class behavior, interpretability, and confidence reliability. Future work could include training on larger and more diverse datasets, using transformer-based vision models, adding ensemble learning, improving calibration, and deploying the system as a real-time or lightweight inference application.